RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2019- 2020 (Odd Semester)

Innovative Teaching Method Execution

UNIT I Introduction

Degree, Semester& Branch: V Semester B.E.EEE
Course Code & Title: EE8591 Digital Signal Processing
Name of the Faculty member: Mrs.G.Gnana Priya
Name of the Topic: Classification of Systems
Name of the Innovative Practice: Mind Map
Date & Duration: 09.07.2019 & 15 Minutes

Description of Mind Map Activity:

A mind map is a graphical way to represent ideas and concepts. It is a visual thinking tool that helps structuring information, helping you to better analyze, comprehend, synthesize, recall and generate new ideas. It is an easy way to brainstorm thoughts organically without worrying about order and structure. It allows the student to visually structure the ideas which is used to help with analysis and recall.

Goals (Learning Outcomes):

- The students will be able to compare the various classifications of systems.

Use of appropriate methods:

Justification for choosing the topic using mind map activity:

Since Classification of Systems is an important topic in concept wise and university exam point of view, I have chosen this topic and the reason for choosing mind map activity is while recollecting the concepts and hints learnt in the classroom, the students will be able to reproduce the concept in the exam.

Effective presentation

Details of the Implementation:

The students were randomly called and they were asked to write the topic they had learnt in the classroom as a pictorial representation in the white board.

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

Due to this activity, the students were able to recall the topics they have learnt in the classroom and they were also able to reproduce it in the exam. In the Internal Assessment Test-I most of the students attended the questions under this topic and scored good marks.

Reflective Critique:

- **Benefits of conducting this activity:**

  The students gave oral feedback that now they got a clear idea about classification of systems, and they asked me to conduct more such activities like this.

- **Challenges:**

  I planned to conduct this activity within 10 minutes. But it took 15 minutes to finish this activity, since few students got hesitate to come to the board and write, I made those students to come forward. So it took 5 minutes extra time to finish this activity.

Samples images of the activity:

Course Outcome	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	2	2	2	-	-	-	-	1	-	2	-	2

CO1: The students will be able to analyze the various properties of signals and systems , their mathematical representation and the various types of sampling and quantization.

References:

1. https://ocw.mit.edu/resources/res-6-007-signals-and-systems-spring-2011/lecture-notes
2. John G. Proakis and D.G. Manolakis, 'Digital Signal Processing Principles, Algorithmsand Applications', Pearson Education, New Delhi, PHI. 2003.
3. Tarun Kumar Rawat, 'Digital Signal Processing', Oxford.

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<u>Innovative Teaching Method Execution</u>

<u>UNIT II Discrete Time System Analysis</u>

Degree, Semester& Branch: V Semester B.E.EEE
Course Code & Title: EE8591 Digital Signal Processing
Name of the Faculty member: Mrs.G.Gnana Priya
Name of the Topic: Convolution
Name of the Innovative Practice: Sorting strips
Date & Duration: 09.08.2019 & 20 Minutes

Description of Mind Map Activity:

Sorting strips activity involves the students to arrange a particular concept or steps in order. This is a group activity where the students discuss the concept and arrange the strips according to the steps involved in the particular topic or concept.

<u>Goals (Learning Outcomes):</u>

- The students will be able to analyse the various steps involved in convolution.

<u>Use of appropriate methods:</u>

Justification for choosing the topic using sorting strips activity:

In general, students will participate eagerly in group activities rather than individual activities. In student point of view since they feel this topic (Convolution) as hard, I planned this activity for this topic to make them feel easy.

<u>Effective presentation</u>

Details of the Implementation:

The strength of the class on that particular day was 42. So I thought of dividing the class in to 7 groups each containing 6 members. The students were asked to be seated in two desks accordingly. Then each group was given some pieces of paper where the procedure for doing convolution is written in each paper but not in order. Each group has to arrange the pieces of paper in correct order to perform convolution operation. The students actively participated in this activity.

<u>Significance of Results:</u>

Assessment of Effectiveness/Success of the Activity:

After conducting this activity, the students were able to recall the topics they have learnt in the classroom and they were also able to reproduce it in the exam. In the Internal Assessment Test - II most of the students attended the questions under this topic and scored good marks.

Reflective Critique:

- **Benefits of conducting this activity:**

  Usually the students don't attend this topic convolution sum since it is a lengthy procedure. But after conducting this activity they got a clear idea about this topic and the collaborative learning skills of the students got increased.
- **Challenges faced in conducting the activity:**

  I planned to conduct this activity in 15 minutes. But it took around 20 minutes to complete since I mentioned numbers in each paper the students got a little confusion whether to arrange the numbers or to arrange the papers based on the sentences written in each paper. Then I clarified their doubt. So it took some 10 minutes extra time to conduct this activity.

Samples images of the activity:

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	2	2	2	-	-	-	-	1	-	2	-	2

CO2: The students will be able to Analyze discrete time LTI(Linear Time Invariant) systems using Z transform and DTFT(Discrete Time Fourier Transform).

References:

1. Poorna Chandra S, Sasikala. B , Digital Signal Processing, Vijay Nicole/TMH,2013.
2. John G. Proakis and D.G. Manolakis, 'Digital Signal Processing Principles, Algorithmsand Applications', Pearson Education, New Delhi, PHI. 2003.
3. Tarun Kumar Rawat, 'Digital Signal Processing', Oxford.

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UNIT III Discrete Fourier Transform & Computation

Degree, Semester& Branch: V Semester B.E.EEE
Course Code & Title: EE8591 Digital Signal Processing
Name of the Faculty member: Mrs.G.Gnana Priya
Name of the Topic: Properties of DFT
Name of the Innovative Practice: Zero Minute Speech
Date & Duration: 16.08.2019 & 15 Minutes

Description of Zero Minute Speech Activity:

Zero minute speech activity makes the student to come forward and share the things they have learnt during that particular class. Due to this activity the student will be able to recall the topic learnt in the classroom and they can reproduce it in the exam.

Goals (Learning Outcomes):

- The students will be able to compare DFT(Discrete Fourier Transform) and DTFT(Discrete Time Fourier Transform)

Use of appropriate methods:

Justification for choosing the topic using Zero minute speech activity:

In general, the students won't study the properties of Fourier series and in the exam also they prefer to choose problems rather than choosing properties of Fourier Series. But if they attend properties they can score full marks when compared to solving problems because there may be a chance to do mistake while solving problems. To change the mentality of the students and to make the students attend this topic in the exam, I have chosen this topic and conducted Zero Minute Speech Activity.

Effective presentation

Details of the Implementation:

The students were asked to recall the topic they had learnt in the classroom and they were randomly called to give speech for few minutes about the Properties of DFT.

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

Due to this activity, the students were able to recall the topics they have learnt in the classroom and they were also able to reproduce it in the exam.

Reflective Critique:

- **Benefits of conducting this activity:**

  By conducting this activity the active learning skills of the students got increased and it was easy for them to recollect the points learnt in the classroom. They will also be able to reproduce the concepts in the exam.

- **Challenges:**

  I planned to conduct this activity within 5 minutes. But it took around 15 minutes to conduct this activity, since few students hesitate to come to the dias and share the concept. I changed their mentality and brought them to the stage to share the concept. Only very few students got chance to come and share their ideas since the time duration for conducting this activity is limited. Next time while conducting this activity, I thought of calling the students who have not participated in the previous activity (to give a chance for them).

Samples images of the activity

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	2	2	2	-	-	-	-	1	-	2	-	2

CO3: The students will be able to apply the concepts of DFT(Discrete Fourier Transform) and FFT(Fast Fourier Transform) to solve Electrical Engineering Problems.

References:

1. Poorna Chandra S, Sasikala. B , Digital Signal Processing, Vijay Nicole/TMH,2013.
2. John G. Proakis and D.G. Manolakis, 'Digital Signal Processing Principles, Algorithms and Applications', Pearson Education, New Delhi, PHI. 2003.
3. Tarun Kumar Rawat, 'Digital Signal Processing', Oxford.

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<u>UNIT IV Design of Digital Filters</u>

Degree, Semester& Branch: V Semester B.E.EEE
Course Code & Title: EE8591 Digital Signal Processing
Name of the Faculty member: Mrs.G.Gnana Priya
Name of the Topic: IIR Realization
Name of the Innovative Practice: NPTEL Video
Date & Duration: 26.09.2019 & 20 Minutes
ICT Tool Used: Projector

Description of NPTEL Video:

In this NPTEL video, the cascade form of realization and parallel form of realization was well explained.

<u>Goals (Learning Outcomes):</u>

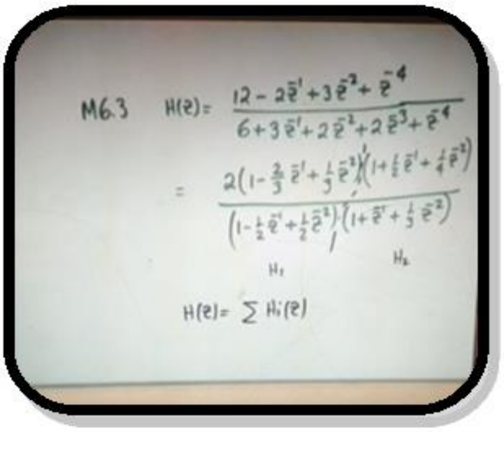
The students will be able to realize the IIR(Infinite Impulse Response) filters.

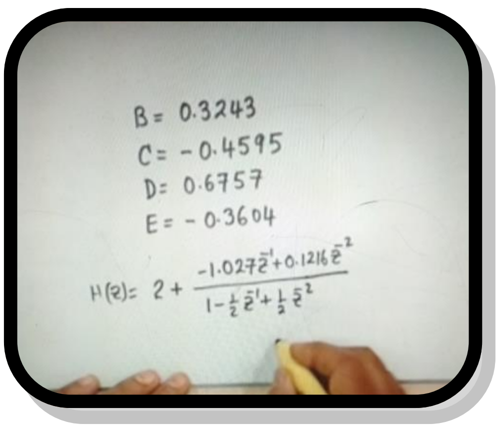
<u>Use of appropriate methods:</u>

Justification for choosing the topic using Zero minute speech activity:

Instead of normal teaching of this topic IIR realization in the class, if the students watch this video, they will get a clear idea about how the cascade and parallel form of realization of IIR filers.

Samples images of the activity:





Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	2	2	2	-	-	-	-	1	-	2	-	2

CO4: The students will be able to design Finite Impulse Response (FIR) and Infinite Impulse Response (IIR) digital filters.

References:

1. NPTEL - Digital Signal Processing – IIR Realizations- Lecture No.: 29
2. John G. Proakis and D.G. Manolakis, 'Digital Signal Processing Principles, Algorithmsand Applications', Pearson Education, New Delhi, PHI. 2003.
3. Tarun Kumar Rawat, 'Digital Signal Processing', Oxford.

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UNIT V Digital Signal Processors

Degree, Semester& Branch: V Semester B.E.EEE
Course Code & Title: EE8591 Digital Signal Processing
Name of the Faculty member: Mrs.G.Gnana Priya
Name of the Topic: Architecture of DSP Processors
Name of the Innovative Practice: Flipped Classroom
Date & Duration: 01.10.2019 & 50 Minutes

Description of Flipped Classroom Activity:

Flipped Classroom is a pedagogical approach in which the conventional notion of classroom based learning is inverted, so that students are introduced to the learning material before class, with classroom time then being used to deepen understanding through discussion with peers and problem solving activities.

Goals (Learning Outcomes):

- The students will be able to analyse and explain the architectural features of digital signal processor

Use of appropriate methods:

Justification for choosing the topic using Flipped Classroom activity:

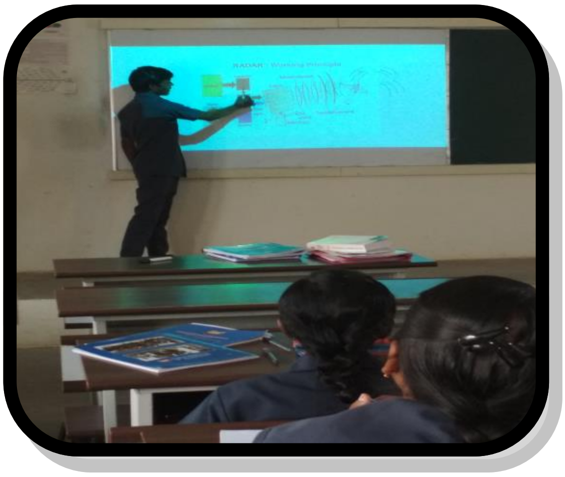
The architecture of DSP Processors is easy to understand and learn. So I planned to conduct flipped classroom activity for this topic.

Effective presentation

Details of the Implementation:

The students were given the materials one week before the conduct of this activity and they were asked to prepare for the presentation. Then on the day of the activity, the assigned students for this activity were asked to present the topic and their doubts were also cleared during that presentation.

<u>Sample Images of the Activity:</u>



Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	2	2	2	-	3	-	-	2	2	2	-	1

CO5: The students will be able to explain the architectural features, addressing formats and the various functional modes of DSP Processors.

References:

1. S.Salivahanan, Digital Signal Processing, Mc Graw Hill.
2. https://ocw.mit.edu/resources/res-6-008-digital-signal-processing-spring-2011/video-lectures/lecture-1-introduction